

SKYPULSE

Airline Operational Performance Report

What causes flight delays and cancellations, and how can operations improve?

Prepared by Malaak Aoughlis

July 2026

Executive Summary

SkyPulse analyses U.S. domestic flight operations in 2025. The main result is that reliability is not evenly distributed. Airlines, airports, routes and departure times show clear performance differences. The strongest operational opportunity is to reduce delay propagation during the day.

300K Flights analysed	77.9% On-time rate	8.3 min Average arrival delay	1.5% Cancellation rate
---	--	---	--

Main conclusion:
morning flights were much more reliable than evening flights. Late-arriving aircraft was the largest recorded delay cause, showing that one disruption can affect later flights.

In simple words

Flights are more reliable early in the day. As delays accumulate, later flights become less punctual. Better aircraft rotation and turnaround planning can improve performance.

Simple definitions

- On-time flight:

a completed flight arriving less than 15 minutes after its scheduled time.
- Average arrival delay:

the average difference between actual and scheduled arrival time.
- Cancellation rate:

cancelled flights as a percentage of all scheduled flights.
- Delay propagation:

a delay that continues into later flights through the same aircraft rotation.

1. Overall Flight Performance

The analysis contains 300,000 flights. 294,803 flights were completed, 4,404 were cancelled and 793 were diverted. The completed-flight on-time rate was 77.9%.

More than three quarters of completed flights arrived on time

Flight status distribution

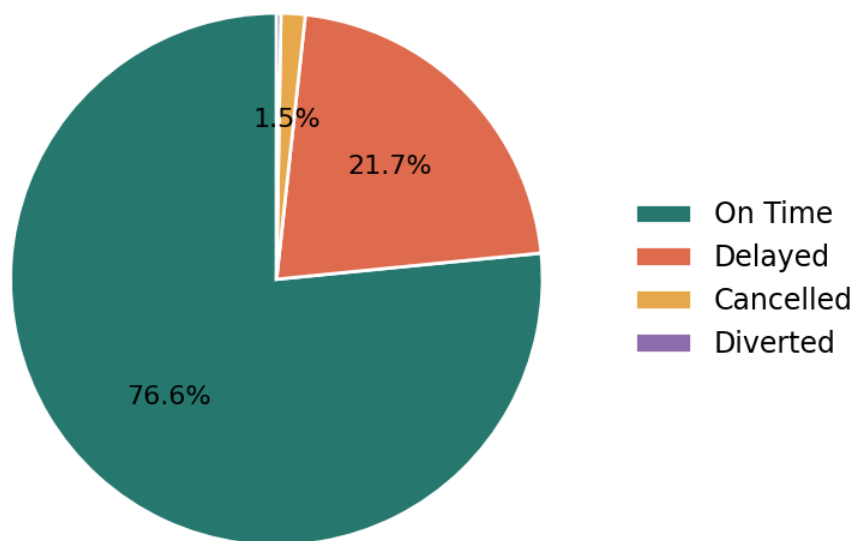


Figure 1. Share of flights by final operational status.

Key result:

229,704 completed flights arrived on time. Average arrival delay was 8.3 minutes and average departure delay was 13.3 minutes.

What this means

The overall rate is useful, but it hides important differences between airlines, airports, routes and departure periods. Managers should monitor both the total result and the weak operational areas.

2. Monthly Performance

Performance changed during the year. Summer disruption was the clearest seasonal risk, while September recorded the strongest result in the sample.

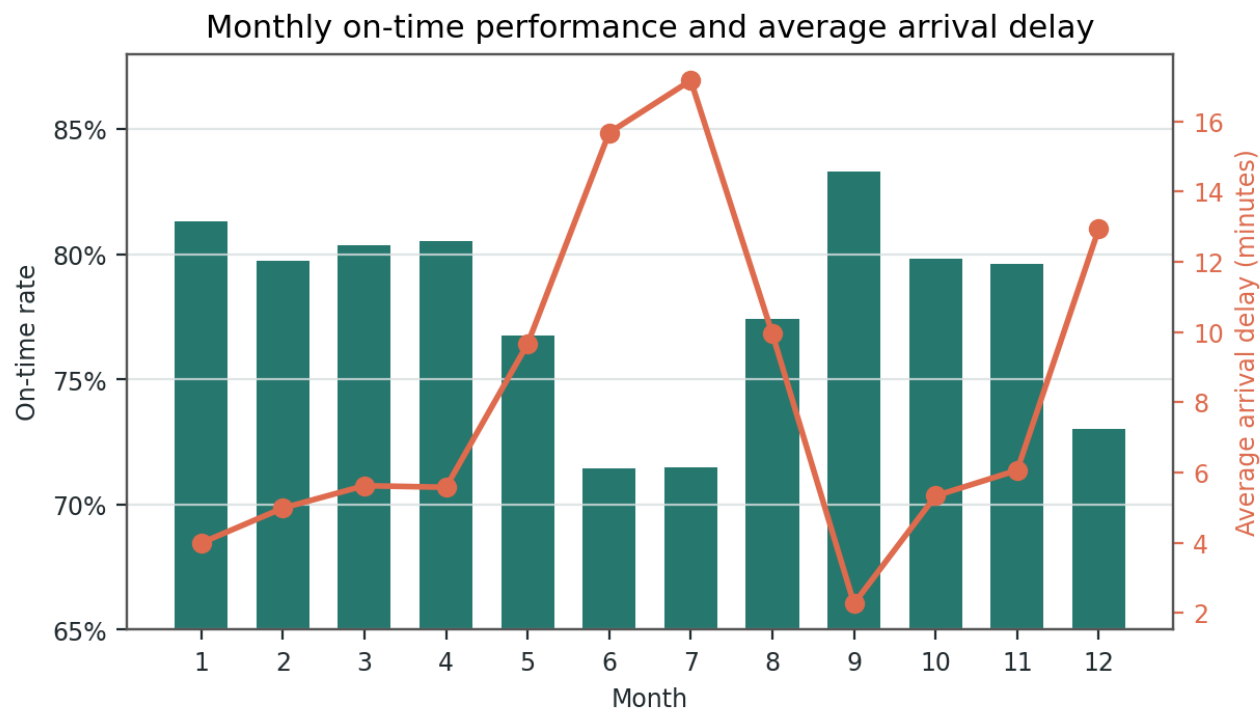


Figure 2. Monthly on-time arrival rate and average arrival delay in 2025.

Seasonal result:

September had the best on-time rate at 83.3%. June and July were both close to 71.5%, and July reached 17.2 minutes of average arrival delay.

What this means

Airlines and airports should prepare additional operational capacity for difficult months. Staffing, turnaround resources and disruption plans should be strengthened before summer demand and weather pressure increase.

3. Airline Performance

Airlines did not achieve the same level of punctuality. The comparison below uses completed flights and measures arrival within 15 minutes of schedule.

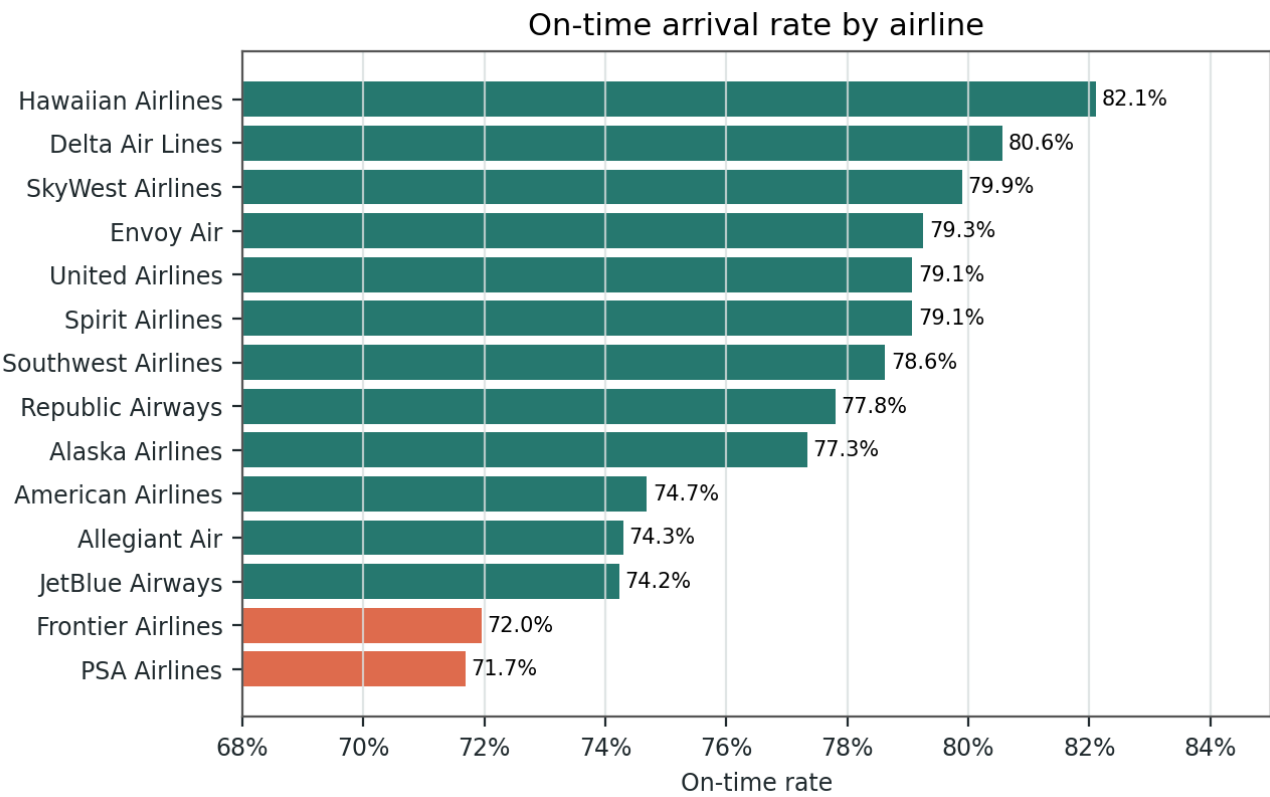


Figure 3. On-time arrival rate for each reporting airline.

Airline result:

Hawaiian Airlines had the highest on-time rate at 82.1%, followed by Delta at 80.6%. PSA and Frontier were close to 72%.

What this means

Airline performance should be monitored together with flight volume, route network and airport exposure. Low punctuality can reflect both internal operations and the difficulty of the network being served.

4. Airport Performance

Some origin airports generated more operational risk than others. A minimum of 1,000 sampled departures was used to avoid ranking very small airports as the best or worst.

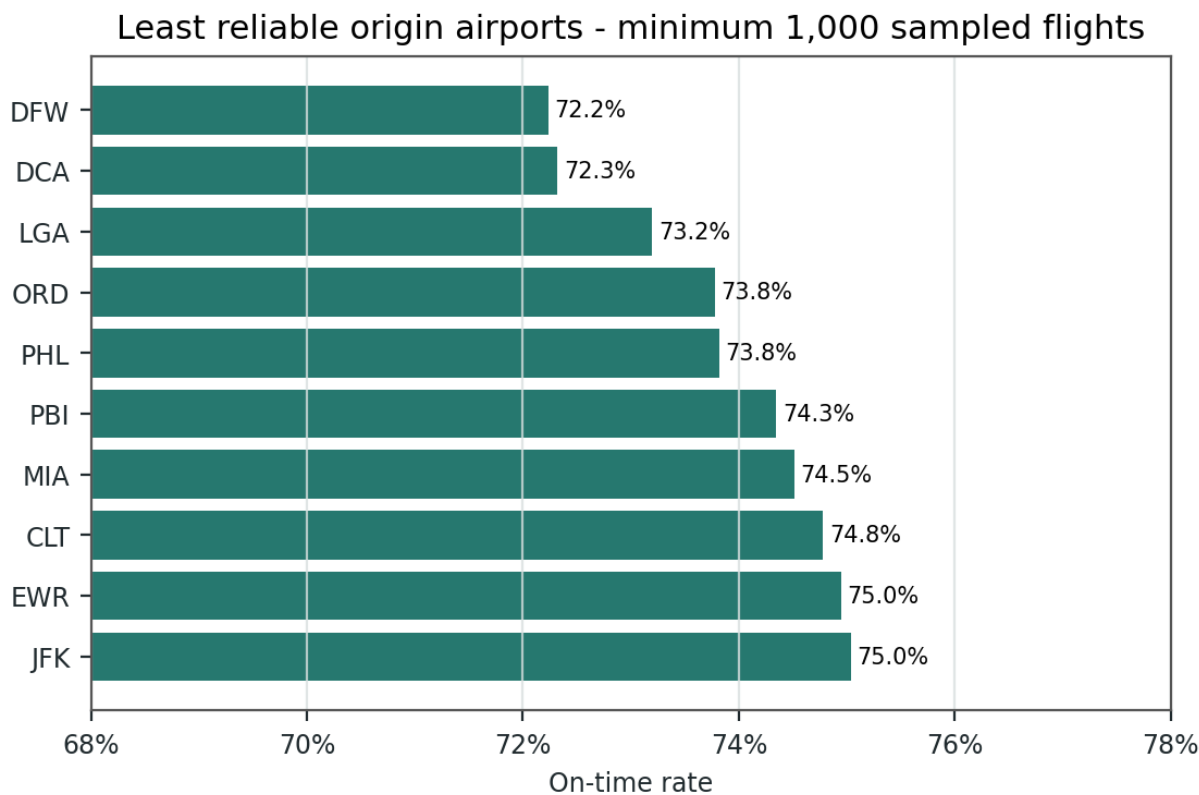


Figure 4. Lowest on-time origin airports after applying a minimum-volume filter.

Airport risk:

DFW had a 72.2% on-time rate. DCA, LGA and ORD were also among the least reliable higher-volume origin airports in the sample.

What this means

High-risk airports need targeted turnaround planning, better disruption communication and additional operational buffer. Airport comparisons should always include a minimum flight-volume rule.

5. Route Performance

Route-level analysis shows where delays occur repeatedly between the same origin and destination. A minimum of 250 sampled flights was used for a more reliable comparison.

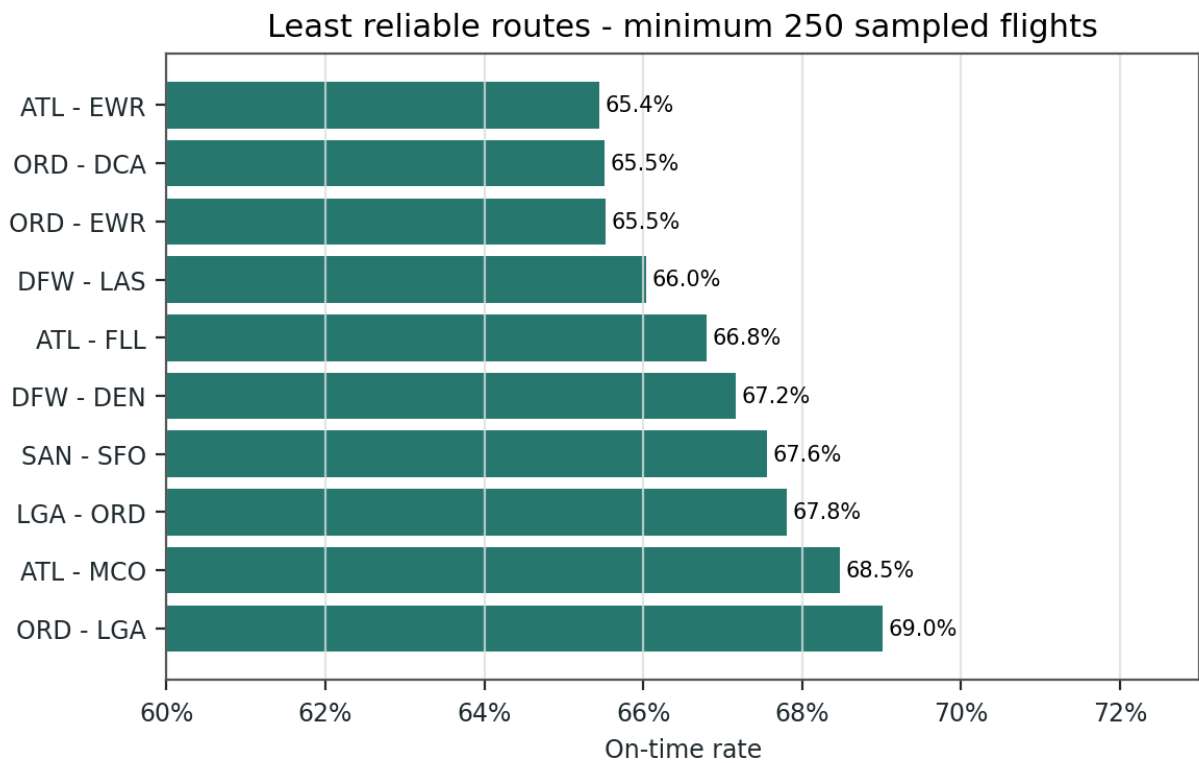


Figure 5. Lowest on-time routes after applying a minimum-volume filter.

Route risk:

ATL-EWR, ORD-DCA and ORD-EWR had on-time rates close to 65.5%. These routes combine repeated operational pressure with weaker reliability.

What this means

Repeatedly unreliable routes should receive schedule reviews, targeted buffer and aircraft-rotation monitoring. The goal is to fix recurring patterns rather than react to each delayed flight separately.

6. Departure Time and Delay Propagation

Departure time was one of the clearest performance factors. Flights became less reliable as the day progressed, supporting the delay-propagation hypothesis.

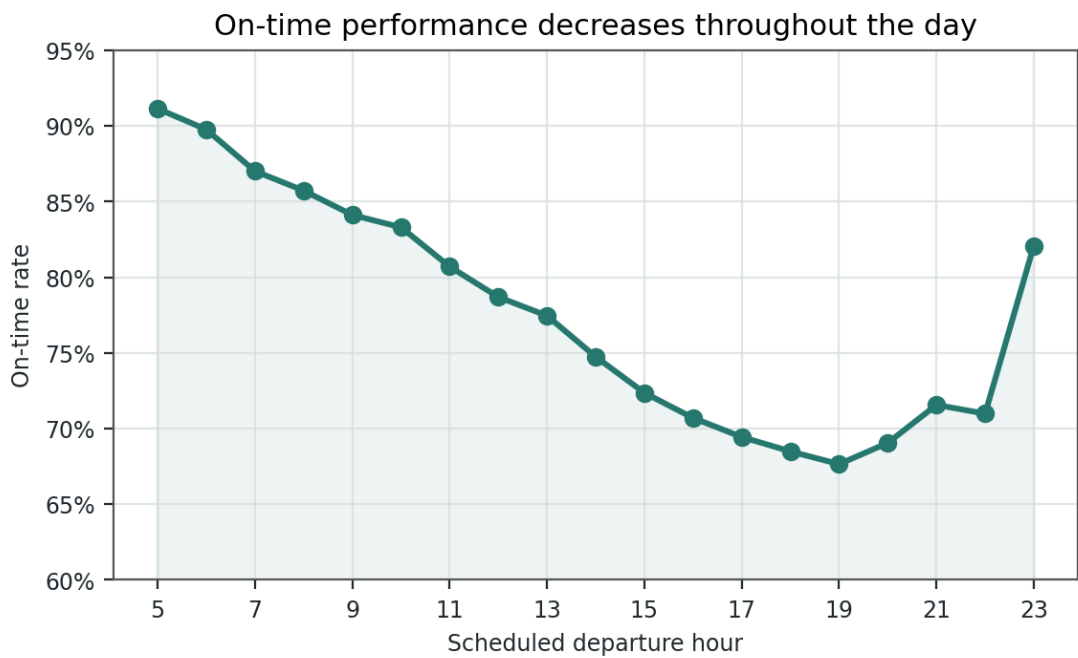


Figure 6. On-time arrival rate by scheduled departure hour.

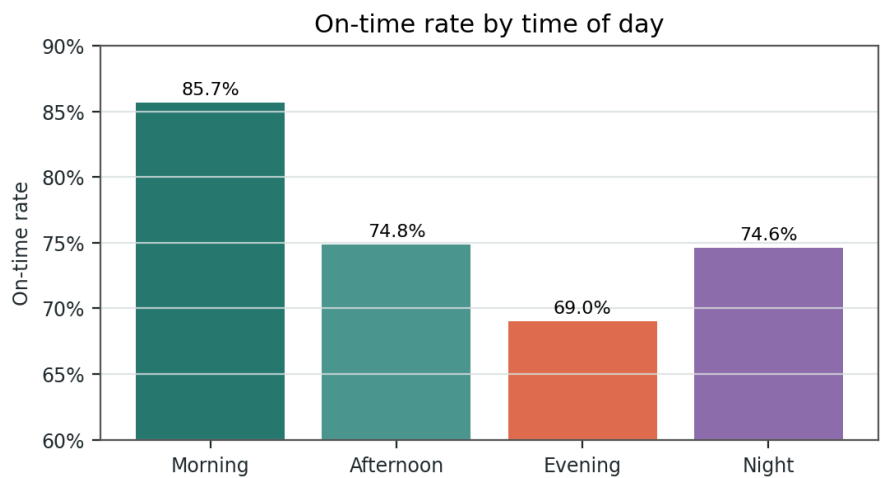


Figure 7. On-time arrival rate by time-of-day group.

What this means

Morning flights were 85.7% on time, compared with 69.0% in the evening. Protecting early schedules and intervening quickly on late inbound aircraft can reduce the daily accumulation of delays.

7. Delay Causes and Cancellations

Delay-cause minutes identify the operational source of major delays. Cancellation rates add another view of disruption severity by airline.

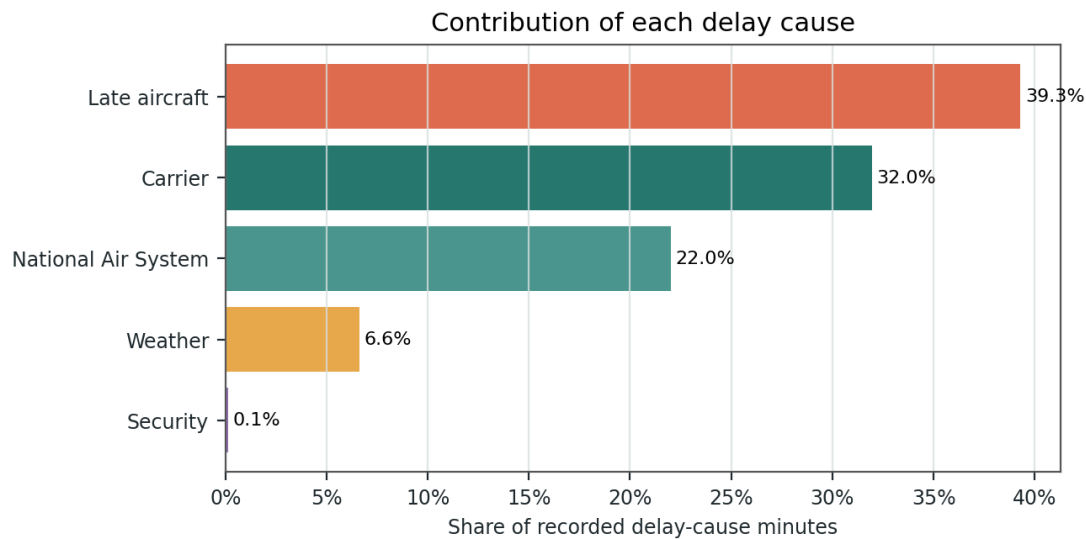


Figure 8. Share of recorded delay-cause minutes.

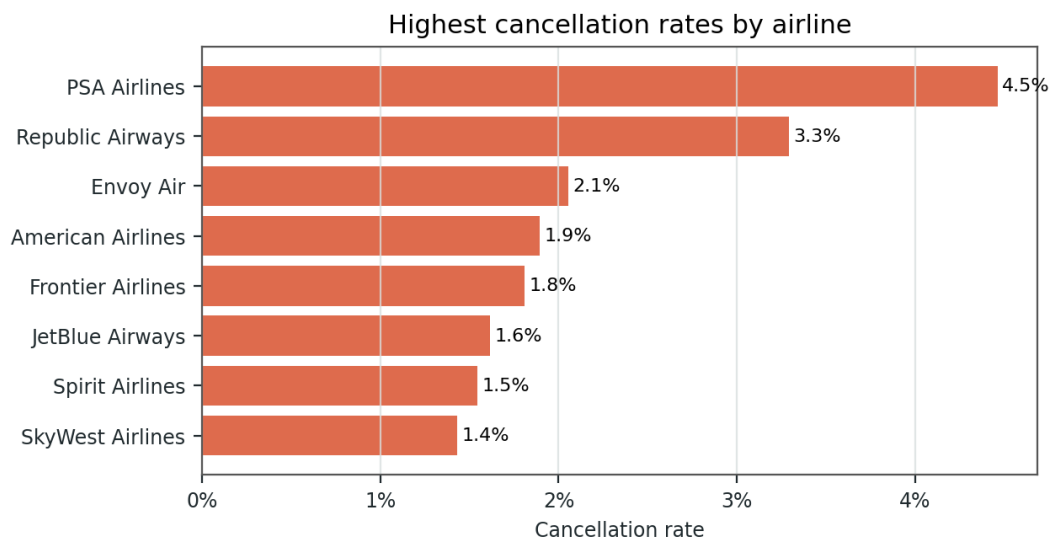


Figure 9. Airlines with the highest cancellation rates.

Main operational risk:

Late-arriving aircraft represented 39.3% of recorded delay-cause minutes. PSA had the highest airline cancellation rate at 4.5%.

What this means

Aircraft rotations are a major improvement opportunity. Better turnaround control and earlier recovery decisions can prevent one late arrival from creating several later delays.

8. Business Recommendations

1. Prioritise early-morning operations

Protect morning schedules and place critical flights earlier when possible.

2. Reduce late-aircraft propagation

Monitor inbound aircraft and intervene before the next departure is affected.

3. Add targeted schedule buffer

Use extra turnaround time on high-risk routes, airports and aircraft rotations.

4. Prepare for difficult months

Increase staffing and recovery resources before June, July and December.

5. Improve weak airports and routes

Create specific action plans for repeated operational problems.

6. Use fair performance comparisons

Apply minimum-flight filters and consider network size when ranking performance.

Recommended priority order

- First: protect morning operations and identify late inbound aircraft early.
- Second: review high-risk aircraft rotations, airports and routes.
- Third: prepare additional resources for the weakest months.
- Fourth: build regular monitoring for delay propagation and cancellations.

9. Analysis Limitations

Important context

- The dashboard uses a balanced sample of 300,000 flights, not the complete 2025 population.
- Each month contains 25,000 sampled flights even though real monthly traffic volumes are different.
- The data covers U.S. domestic reporting carriers and does not represent all global aviation.
- Delay causes describe reported operational minutes but do not explain every business or technical cause.
- Airport and route totals are affected by traffic volume, so minimum-volume filters are necessary.
- Weather observations were not joined to individual flights in this version of the project.

How to use these findings

- Use the results to guide schedule, turnaround and route reviews.
- Test operational changes and monitor on-time rate, delay minutes and cancellation rate over time.
- Use the complete population and current operational data before making major planning decisions.
- Add weather and aircraft-rotation data for a deeper root-cause analysis.

10. Final Conclusion

SkyPulse shows that flight delays are not evenly distributed. The main operational problem is not only the total number of delayed flights, but the way disruption accumulates across aircraft rotations, airports, routes and departure periods.

Morning flights were significantly more reliable than evening flights, and late-arriving aircraft was the largest recorded delay cause. This means that airlines can improve performance by protecting early schedules, monitoring inbound aircraft and adding targeted buffer where repeated problems occur.

Final message

Airlines and airports can improve reliability without simply adding more capacity. Better timing, turnaround control and faster action on delay propagation can protect the flights already scheduled.

Project source

Bureau of Transportation Statistics - Airline On-Time Performance

Report basis: the project README, Python cleaning notebook, exploratory data analysis notebook and Power BI dashboard. The charts in this report were rebuilt from the cleaned 300,000-flight dataset for a clear executive presentation.